**16** When a sample of ideal monatomic gas has 1000 J of heat added to it and it does 500 J of work, its temperature changes by $\Delta T_1$. When twice the amount of heat is added to it and it does the same amount of work, its temperature changes by $\Delta T_2$.

What is the ratio $\frac{\Delta T_1}{\Delta T_2}$?

**A** 0.20 **B** 0.33 **C** 0.50 **D** 0.60